

3.1 Electronegativity homework — answers

CIE 9701 AS Chemistry · 11 multiple-choice + 5 written marks

Do not issue this file to students.

Section A — Multiple choice

Quick answers

Q	Ans.	Q	Ans.	Q	Ans.
A1	A	A5	B	A9	C
A2	C	A6	B	A10	B
A3	C	A7	D	A11	C
A4	C	A8	A		

Worked explanations

A1. A

2022 MJ Paper 12 Q19

Across Period 3 from Na to Si the nuclear charge rises while the outer electrons stay in the same shell, so the atomic radius *decreases* and the electronegativity *increases*. The correct graph must therefore slope downwards, with Si (highest electronegativity, smallest radius) at the top-left and Na (lowest electronegativity, largest radius) at the bottom-right.

B: the elements are in the right order on the radius axis, but Al is placed above Si — Si has the highest electronegativity of the four, so Si must be the highest point.

C: shows electronegativity *increasing* with atomic radius (Na at the top, Si at the bottom) — the trend is the wrong way round.

D: swaps Al and Si on the radius axis (Al appears to the left of Si), but the correct radius order is $\text{Si} < \text{Al} < \text{Mg} < \text{Na}$.

A2. C

2023 FM Paper 12 Q22

Use the two radii to identify the elements. V has a large atomic radius (0.186 nm) but a much smaller ionic radius (0.095 nm) — it is a metal that has *lost* its outer electron. This matches Na (Na 0.186 nm, Na^+ 0.095 nm). Z has a small atomic radius (0.099 nm) but a *larger* ionic radius (0.181 nm) — it is a non-metal that has *gained* an electron. This matches Cl (Cl 0.099 nm, Cl^- 0.181 nm). So $V = \text{Na}$ and $Z = \text{Cl}$.

A: Na^+ has two shells (2,8) but Cl^- has three (2,8,8), so they do *not* have the same number of full shells.

B: Z is chlorine, which forms a *negative* ion, Cl^- .

C (correct): Cl is far more electronegative than Na (3.2 vs 0.9).

D: Na has 1 outer electron and Cl has 7, so V has *fewer* outer electrons than Z.

A3. C

2022 FM Paper 12 Q18

Electronegativity *increases* across Period 3 (nuclear charge increases, shielding stays roughly constant, radius falls). The oxides change from *basic* (Na_2O , MgO) through *amphoteric* (Al_2O_3) to *acidic* (SiO_2 , P_4O_{10} , SO_3 , Cl_2O_7). Only row C has both trends correct.

A / B: both state that electronegativity *decreases* across the period — the wrong direction.

D: has the correct electronegativity trend but the acid–base order reversed.

A4. C

2023 MJ Paper 13 Q17

Astatine is below iodine in Group 17. Going down the group the atomic radius and the shielding increase, so the nucleus attracts the bonding electrons less and the electronegativity *decreases*. The larger, more electron-rich molecule also has stronger van der Waals' forces between molecules, so astatine is *less volatile* (higher boiling point) than iodine.

A / B: both say astatine is *more* electronegative — wrong for a heavier halogen.

D: correct lower electronegativity, but says astatine is *more* volatile; volatility decreases down the group.

A5. B

2025 MJ Paper 11 Q20

Electronegativity increases across $\text{Na} \rightarrow \text{Mg} \rightarrow \text{Al} \rightarrow \text{Si}$. First ionisation energy also increases *overall*, but not in atomic-number order: Mg ($3s^2$, a full subshell) has a *higher* first ionisation energy than Al ($3p^1$, easier to remove). So the graph must show the points in the order $\text{Na} < \text{Al} < \text{Mg} < \text{Si}$ along the ionisation-energy axis, while the electronegativity order is $\text{Na} < \text{Mg} < \text{Al} < \text{Si}$ — i.e. Mg lies above Al, and Si is both furthest right and highest.

A: places Mg before Al on the ionisation-energy axis, giving Al a higher first ionisation energy than Mg — this ignores the $3s^2 / 3p^1$ difference.

C / D: show electronegativity *decreasing* as ionisation energy increases; in fact the two trends rise together across the period.

A6. B

2025 ON Paper 14 Q17

Electronegativity increases $\text{Mg} < \text{Al} < \text{Si} < \text{P}$, so the points must appear in that order from left to right. The melting point depends on structure: Mg and Al are metals with similar melting points ($\approx 650\text{--}660^\circ\text{C}$); Si is a giant covalent (macromolecular) solid with a much higher melting point (1414°C); P exists as simple P_4 molecules held by weak van der Waals' forces, so it melts at only 44°C . Graph B is the only one showing Si clearly the highest, Mg and Al close together and much lower, and P the lowest.

A: puts Al above Si — the melting point of Si is far higher than that of Al.

C / D: place the elements in the wrong order on the electronegativity axis (P to the left of Si/Mg), reversing the Period 3 trend.

A7. D

2022 MJ Paper 13 Q5

The difference in electronegativity is smallest for the pair of elements that are closest in electronegativity. Using Pauling values (K 0.8, Li 1.0, Br 3.0, F 4.0): $\Delta E_{\text{N}}(\text{KF}) \approx 3.2$, $\Delta E_{\text{N}}(\text{LiF}) \approx 3.0$, $\Delta E_{\text{N}}(\text{KBr}) \approx 2.1$, $\Delta E_{\text{N}}(\text{LiBr}) \approx 2.0$. Li and Br are the closest pair, so LiBr has the smallest difference.

A / C: fluorine is the most electronegative element, so both KF and LiF give large differences.

B: KBr has the same anion as D, but K is less electronegative than Li, so its difference is larger.

A8. A

2023 MJ Paper 13 Q4

L has the *highest* Pauling value of the seven elements Na–Cl, so L is chlorine. M has the *lowest*, so M is sodium. (1) Chlorine exists as Cl_2 , which contains a covalent Cl–Cl bond — **true**. (2) Cl ($Z = 17$) has a higher atomic number than Na ($Z = 11$) — **true**. (3) Sodium and chlorine form NaCl, a giant ionic lattice — **true**. All three statements are correct, so the answer is A.

B / C / D: each option rejects at least one statement that is in fact correct — the trap is assuming that an element cannot contain covalent bonds (it can, e.g. Cl_2), or that Na and Cl would bond covalently.

A9. C

2022 FM Paper 12 Q7

Identify the elements from their electronegativity: X (1.0) = Na, Y (2.1) = H, Z (4.0) = F. $Z_2 = F_2$: simple molecular, only weak van der Waals' forces — *lowest* melting point. $YZ = HF$: simple molecular with hydrogen bonding — higher than F_2 , but still low. $XZ = NaF$: giant ionic lattice with strong electrostatic forces — *highest* melting point. So increasing melting point is $Z_2 < YZ < XZ$, which is row C.

A / B: put the ionic compound XZ at the lowest end — ionic lattices have by far the strongest forces here.

D: puts YZ (HF) highest; hydrogen bonding between small molecules is much weaker than the electrostatic forces in an ionic lattice.

A10. B

2021 MJ Paper 13 Q6

Ionic bonding requires the *transfer* of electrons from a metal atom to a non-metal atom, i.e. two different elements. A single element contains only one type of atom and cannot form oppositely charged ions, so ionic bonding is *never* found in elements.

A: covalent — H_2 , Cl_2 , diamond.

C: metallic — Na, Fe.

D: van der Waals' forces — between I_2 molecules or noble-gas atoms.

A11. C

2023 MJ Paper 12 Q19

(1) **True** — the ionically bonded oxides Na_2O and MgO are basic and neutralise dilute hydrochloric acid. (2) **False** — this is the trap. $NaCl$ does give a neutral solution, but $MgCl_2$ gives an *acidic* solution (the small, highly charged Mg^{2+} ion polarises water), so not all metal chlorides are neutral. (3) **True** — the two most electronegative elements of the set are Si and P, and $SiCl_4$ and PCl_3/PCl_5 are covalent. Correct statements: 1 and 3 → C.

Section B — Mark scheme clips

B1(a)(i) [1]

Question	Answer	Marks
1(a)(i)	power of an atom to attract electrons to itself	1

B1(a)(ii) [2]

1(a)(ii)	• O lower nuclear charge / lower proton number • O has (one) fewer shell than S / less shielding • greater attraction (for nucleus) in O	2
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B2(b)(ii) [2]

1(b)(ii)	(across a period) • increase in nuclear charge • similar shielding • (so) increase in nuclear attraction for bonding / outer / valence electrons OR bonding / outer / valence electron(s) are more strongly attracted to nucleus Two correct for one mark, three correct for two marks	2
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